FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2021

PAPER – 2

TEST DATE: 04-07-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

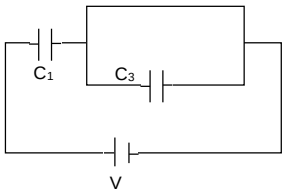
SECTION – A

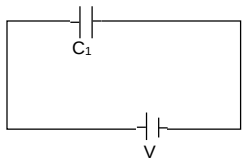
1. ABC

Sol. When $C_2 \to \infty \Rightarrow$ shorted

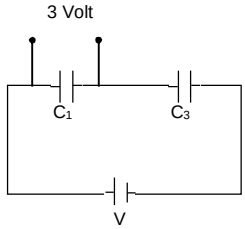
Then the circuit becomes

$\therefore V_1 = V = 12$ volt





When $C_2 = 0$ then circuit becomes



$\therefore V_{C_3} = 12 - 3 = 9V$

Since, C_1 and C_3 are in series, so charge will be same on both the capacitors.

$\therefore C_1 \times 3 = C_3 \times 9 \Rightarrow \dfrac{C_1}{C_3} = 3$

2. B

Sol. $0 = dU + PdV = d(a + bPV) + PdV$

After solving

$PV^{\frac{b+1}{b}}$ = constant

3. ABC

Sol. $V_{IP} = \left(\frac{f}{f-u}\right)^2 \times \frac{36}{\mu} = \left(\frac{-22.5}{-22.5+90}\right) \times \left(\frac{36}{4/3}\right) = 3\,cm/s$

4. AB

Sol. Distance of the target $x = (n)\frac{2\pi m}{qB} v\cos\theta$, where n is an integer

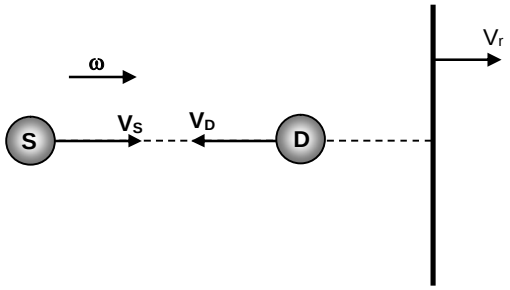
$\frac{nv}{B} = \frac{v_0}{B_0}$

5. BC

Sol. The frequency received by the reflector is

$f_1 = \left(\frac{V + W - V_r}{V + W - V_s}\right) f$

The frequency of the reflected wave received by the detector is



$f_2 = \left(\frac{V + W - V_D}{V + W - V_r}\right)\left(\frac{V + W - V_r}{V + W - V_s}\right) f$

$= \left(\frac{330 - 10 - 10}{330 - 10 + 20}\right)\left(\frac{330 + 10 - 20}{330 + 10 - 30}\right) 527$

$f_2 = \left(\frac{310}{340}\right)\left(\frac{320}{310}\right) \times 527 = 496\,Hz$

The wavelength of the reflected wave received by the detector is

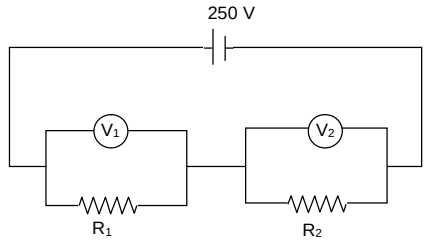
$\lambda_2 = \frac{V - W - V_D}{f_2} = \frac{330 - 10 - 10}{496} = \frac{5}{8} m$

6. BD

Sol. When switch S is open

$V_1 = \frac{250}{10000}(6000) = 150\,V$

$V_2 = \frac{250}{10000}(4000) = 100\,V$



When switch S is closed circuit reduces to shown in the figure.

So both the voltmeters show readings $V_1 = V_2 = \frac{250}{2} = 125\,V$

SECTION – B

7. 8

Sol. Initially $P_{in} = \frac{4T}{r}$, where P_{in} = inside pressure …(i)

$P_{outside} = 0$ (always)

When charge is sprayed, for equilibrium $P_{in}^{'} + \frac{\sigma^2}{2\varepsilon_0} = \frac{4T}{2r}$ ...(ii)

Since, isothermal process

$\therefore P_{in}^{'} = \frac{P_{in}V_i}{V_f} = \frac{P_{in}}{8}$ ...(iii)

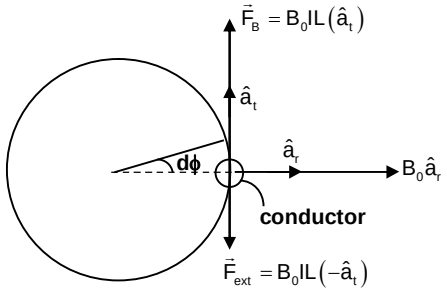
Using equations (i), (ii) and (iii) we have, $\frac{4T}{8r} + \frac{\sigma^2}{2\varepsilon_0} = \frac{4T}{2r} \Rightarrow \sigma = \sqrt{\frac{3T\varepsilon_0}{r}}$

8. 5

Sol. $\vec{F}_B = I\left(\vec{L}\times\vec{B}\right) = I\left(LK\times B_0\hat{a}_r\right) = B_0IL\hat{a}_t$

$\vec{F}_{external} = B_0IL\left(-\hat{a}_t\right)$

Work done by external agent to turn the conductor by 2π radian



$W = \int_0^{2\pi} B_0IL\left(-\hat{a}_t\right).rd\phi\hat{a}_t = -2\pi rB_0IL$

$P = 2\pi rB_0ILN = 0.5\,W$

9. 5

Sol. When the wedge has moved through a distance y, let the point of contact of cylinder have moved down the plane through x (with respect to wedge). Let F be the friction between the cylinder and the plane.

Since there is no horizontal force on 9wedge + cylinder system)

$$M\frac{d^2y}{dt^2}+m\left(\frac{d^2y}{dt^2}-\cos\alpha\frac{d^2x}{dt^2}\right)=0 \qquad ...(i)$$

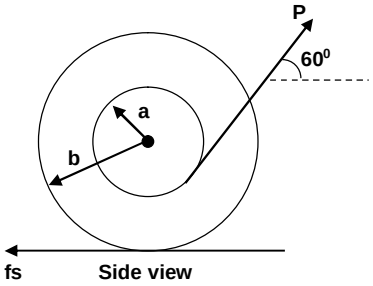
$$\text{Also, } m\left(\frac{d^2y}{dt^2}\cos\alpha-\frac{d^2x}{dt^2}\right)=F-mg\sin\alpha \text{ and } \frac{ma}{2}\frac{d^2x}{dt^2}=\frac{ma^2}{2}\frac{d^2\theta}{dt^2}=Fa, \text{ so that}$$

$$2\frac{d^2y}{dt^2}\cos\alpha-3\left(\frac{d^2x}{dt^2}\right)=-2g\sin\alpha \qquad ...(ii)$$

Equation (i) and (ii) implies acceleration of the centre of the cylinder down the face, and relative to the wedge, is $\frac{2g\sin\alpha(M+m)}{3M+m+2m\sin^2\alpha}=5\ m/s^2$

10. 5

Sol. Assume that the drum rolls away from the force P, so that the friction F is away from P. Then:



$$Mk^2\frac{d^2\theta}{dt^2}=Pa-Fb \text{ and } Mb\frac{d^2\theta}{dt^2}=M\frac{d^2x}{dt^2}=F-P\cos60^0$$

$$M\left(k^2+b^2\right)\frac{d^2\theta}{dt^2}=P\left[a-\frac{b}{2}\right]$$

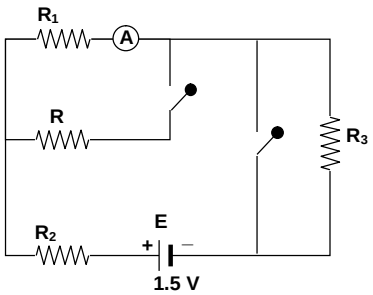
$$\Rightarrow 0.25\times8\frac{d^2\theta}{dt^2}=10(1.5-1)$$

$$\Rightarrow\frac{d^2\theta}{dt^2}=2.5\,rad/s^2$$

and the required acceleration $=b\frac{d^2\theta}{dt^2}=2\times2.5=5\ m/s^2$

11. 6

Sol. With both switches open: $I_A=\frac{E}{R_1+R_2+R_3}$



With both switches closed:
Current through E and R_2

$$I' = \frac{E}{R_2 + \frac{RR_1}{R+R_1}}$$

Current through the ammeter

$$I_A^{'} = \frac{RI'}{R+R_1}$$

$$I_A = I_A^{'}$$

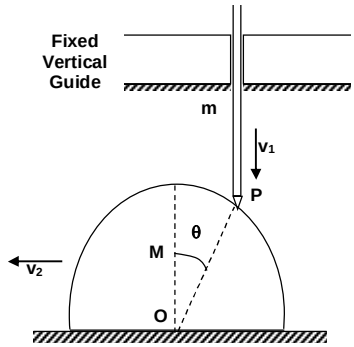
$$R = \frac{R_1R_2}{R_3} = 600\Omega$$

12. 2

Sol. $v_1\cos\theta = v_2\sin\theta$

$v_1 = v_2\tan\theta$

$v_1 = v_2\sqrt{3}$ (given $\theta = 60^0$)



Now using conservation of energy

$$mga(1-\cos\theta) = \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2$$

$\Rightarrow \quad 2ga = 2v_1^2 + 4v_2^2$

$40 = 6v_2^2 + 4v_2^2$

$\Rightarrow \quad 10v_2^2 = 40$

$\Rightarrow \quad v_2 = 2\ m/s$

$\therefore \quad v_1 = 2\sqrt{3}\ m/s$

$$\omega = \frac{v_1\sin\theta + v_2\cos\theta}{a} = 2\ rad/s$$

SECTION – C

13. 00000.80

Sol. $I_{C.B.F} = K\left[A_1^2 + A_2^2 + 2A_1A_2\cos\phi\right] = I_0$

$$KA^2 = \frac{I_0}{16}, \left(A_1 = A, A_2 = 3A \text{ and } \phi = 0^0\right)$$

$$I_0 = \frac{16}{25}K\left[A_1^2 + A_2^2 + 2A_1A_2\cos\phi\right] = 1$$

$$\phi = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda} \times d\sin\theta = \frac{2\pi}{\lambda} \times d\frac{3}{5}$$

So, $\phi = \pi$

$$\text{Hence, } I = \frac{I_0}{4} \times \frac{16}{25} = \frac{5\times16}{100} = 0.80\ W/m^2$$

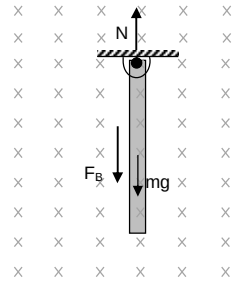
14. 00032.50

Sol. Loss in gravitational potential energy = Gain in Kinetic energy

$$Mg\left[\frac{L}{2}-\frac{L}{2}\cos 60^0\right]=\frac{1}{2}\frac{ML^2}{3}\omega^2$$

$$\omega=\sqrt{\frac{3g}{2L}}=\sqrt{15}\ \text{rad/s}$$

When rod is at its lowest position the forces acting on it are as shown in the diagram.



Magnetic force on the rod

$$F_B=\int dq\left(\vec{v}\times\vec{B}\right)=\int_{x=0}^{x=L}\left(\frac{q}{L}dx\right)\times(\omega X)\times B=\frac{1}{2}qBL\omega$$

$$\therefore N-mg-F_B=m\omega^2\frac{L}{2}$$

$$N=mg+F_B+M\omega^2\frac{L}{2}$$

$$=1\times10+\frac{1}{2}\times1\times\sqrt{15}\times2\sqrt{15}\times1+1\times\left(\sqrt{15}\right)^2\times\frac{1}{2}$$

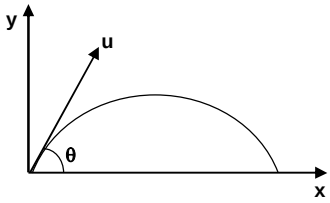
$$=10+15+7.5=32.5N$$

15. 00195.00

Sol. Coordinates of the ball at t = t is

$$x=u\cos\theta t$$

$$y=u\sin\theta t-\frac{1}{2}gt^2$$



and time of light is $\frac{2u\sin\theta}{g}$

Area under trajectory is

$$A=\int_0^x ydx=\int_0^{\frac{2\sin\theta}{g}}\left(u\sin\theta t-\frac{gt^2}{2}\right)u\cos\theta dt$$

$$\therefore A=\frac{2u^4}{3g^2}\sin^3\theta\cos\theta$$

For A to be maximum, $\frac{dA}{d\theta}=0$

$$\Rightarrow\theta=60^0$$

$$\therefore A_{max}=\frac{\sqrt{3}u^4}{8g^2}=194.85\approx195m^2$$

16. 00015.38

Sol. Total work done by gas per cycle, $\Delta W_{cycle}=P_0V_0$

During the process AB

$$\Delta Q_{AB} = nC_v\Delta T = n\frac{3R}{2}\Delta T = \frac{3}{2}V_0\left(2P_0 - P_0\right) = \frac{3}{2}P_0V_0$$

During the process BC

$$\Delta Q_{BC} = nC_P\Delta T = n\frac{5R}{2}\Delta T = \frac{5}{2}\times 2P_0\left(2V_0 - V_0\right) = 5P_0V_0$$

Efficiency of the cycle

$$\eta = \frac{\Delta W_{cycle}}{Q_{supplied}}\times 100 = \frac{P_0V_0}{\left(\frac{3P_0V_0}{2} + 5P_0V_0\right)}\times 100 = \frac{1}{13}\times 100 = 15.38\%$$

17. 00002.50

Sol. $v_0 = \omega h\sin\theta$

$$\Rightarrow \omega = \frac{v_0}{h\sin\theta} = \frac{2v_0}{h} = 2.50 \text{ rad/s}$$

18. 00004.50

Sol. When the ball A leaves the horizontal surface

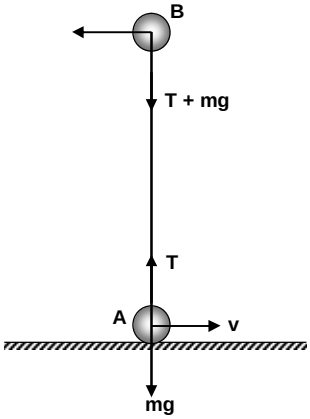
$T = mg$ … (i)

$$T + mg = \frac{m(2v)^2}{\ell}$$

$$2m = \frac{4mv^2}{\ell}$$

$$v = \sqrt{\frac{g\ell}{2}} \quad \text{...(ii)}$$

Now, using conservation of Energy



$$\frac{1}{2}mu^2 = 2\times\frac{1}{2}mv^2 + mg\ell$$

$$u^2 = 2v^2 + 2g\ell$$

$$u^2 = g\ell + 2g\ell$$

$$\Rightarrow u_{min} = \sqrt{3g\ell} = \sqrt{3\times 10\times 0.675} = 4.50 \text{ m/s}$$

Chemistry

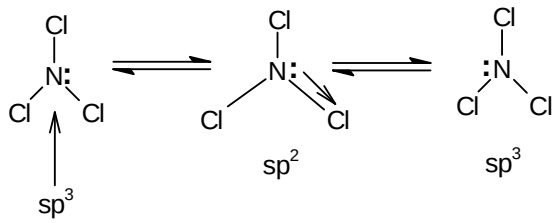
PART – II

SECTION – A

19. ACD

Sol.

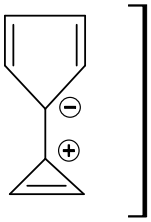
(A)



(electronegative nitrogen become more electronegative because chlorine atoms pull electron density away from nitrogen than methyl group in $N(CH_3)_3$, creating electron defficient N, and destabililising the T.S)

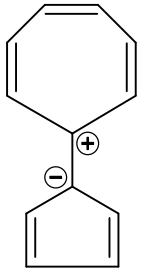
(B) After charge separation due to shifting of π electron

(1)



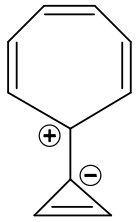
2 aromatic ring stabilised by resonance.

(2)



2 aromatic ring more stable than (1) due to extra resonance

(3)

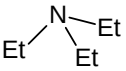


1 aromatic ring

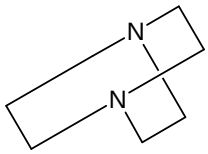
1 anti - aromatic ring required more energy for charge separation

Bond rotation energy is III > I > II
Bond rotation order II > I > III

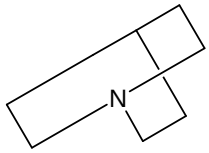
(C)



Least basic
due to amine inversion



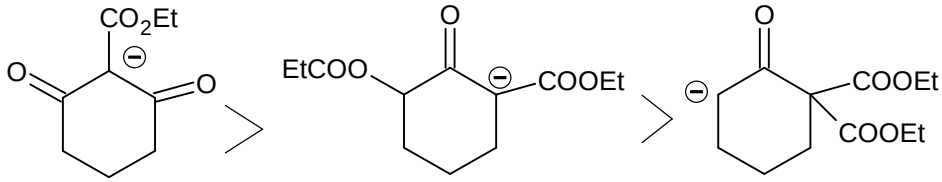
Less basic
due to - I of 2nd N



most basic

No amine inversion

(D) Stability of conjugate base α Acidic nature of conjugate acid



(stability order of C.B.)

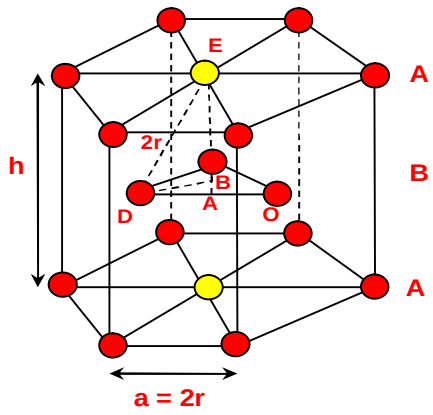
20. ABCD

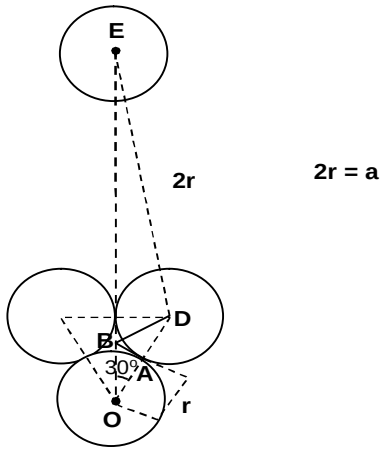
Sol. (A) Co-ordination number in HCP = 12

(B) Volume = Base area $\times$ height (h)

Base area = $6 \times$ area of equilateral triangle

$= 6 \times \frac{\sqrt{3}}{4} a^2$





2r = a

$OA = r$

$\angle AOB = 30^o$

$OB = BD = \frac{2r}{\sqrt{3}}$

$EB = \sqrt{(2r)^2 - \left(\frac{2r}{\sqrt{3}}\right)^2}$

$= 2r\sqrt{\frac{2}{3}} = \frac{h}{2}$

$h = 4r\sqrt{\frac{2}{3}}$

Now, volume = h $\times$ area of base

$= 4r\sqrt{\frac{2}{3}} \times 6 \times \frac{\sqrt{3}}{4} a^2$

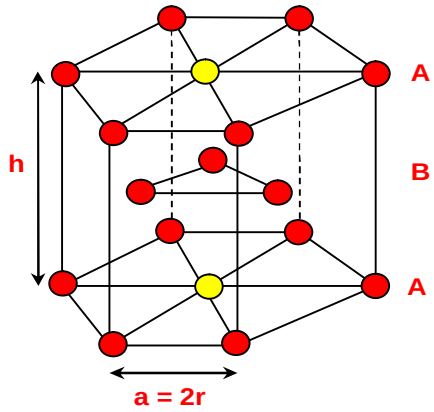
$= 4r\sqrt{\frac{2}{3}} \times 6 \times \frac{\sqrt{3}}{4} (2r)^2$

$= 24\sqrt{2}\ r^3$

$(C) \Rightarrow$ Packing fraction $= \dfrac{6 \times \frac{4}{3}\pi r^3}{\text{volume of unit of unit cell}}$

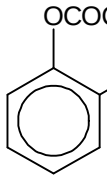
$= \dfrac{6 \times \frac{4}{3}\pi r^3}{24\sqrt{2}\, r^3} = \dfrac{\pi}{3\sqrt{2}}$

(D) $\Rightarrow Z = 12\ (\text{at corner}) \times \dfrac{1}{6} + 2(\text{face centres})\dfrac{1}{2} + 3(\text{in the body}) = 6$



Z = No. of atoms per unit cell.

21. ABCD

Sol. (A) Due to stabilization of intermediate
(B) Less steric hindrance
(C) Fact
(D) A is  Aspirin, an antipyretic.

22. ABC

Sol. Molten Zn and Pb are immiscible liquid and Ag is more soluble in molten Zn than molten Pb.

23. ABC

Sol. $ZnCl_2 + 2NaOH \rightarrow Zn(OH)_2 \xrightarrow{2NaOH} Na_2ZnO_2 + 2H_2O$

$Na_2ZnO_2 + H_2S \rightarrow ZnS\downarrow + 2NaOH$

$Na_2ZnO_2 + 4CH_3COOH \rightarrow (CH_3COO)_2 Zn + 2CH_3COONa + 2H_2O$

$2(CH_3COO)_2 Zn + K_4[Fe(CN)_6] \rightarrow Zn_2[Fe(CN)_6] + 4CH_3COOK$

$3Zn_2[Fe(CN)_6] + K_4[Fe(CN)_6] \rightarrow 2Zn_3K_2[Fe(CN)_6]_2 \downarrow$ White

24. ABCD

Sol. (A) Unacceptable because it is infinite over a finite region
(B) Unacceptable because it is not single valued
(C) Unacceptable because it is not continuous
(D) Unacceptable because its slope is discontinuous

SECTION – B

25. 4

Sol. **Statement – 1** Singlet carbene has zero unpaired electron hence spin multiplicity

$= 2s + 1$ $\Rightarrow$ s = total spin

$= 1$ $\Rightarrow s = \frac{n}{2}, \quad n = 0$

Statement – 2 $H^{\ominus}$,no. of electron = 2

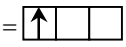
According to (n + ℓ) rule

$1s^2$ $2s^0$ $2p^0$

$\downarrow$

ground state 1st excited state 2nd excited state

Hence

$1s^1 \ 2s^o \ 2p^1 \Rightarrow 2p^1 =$ 

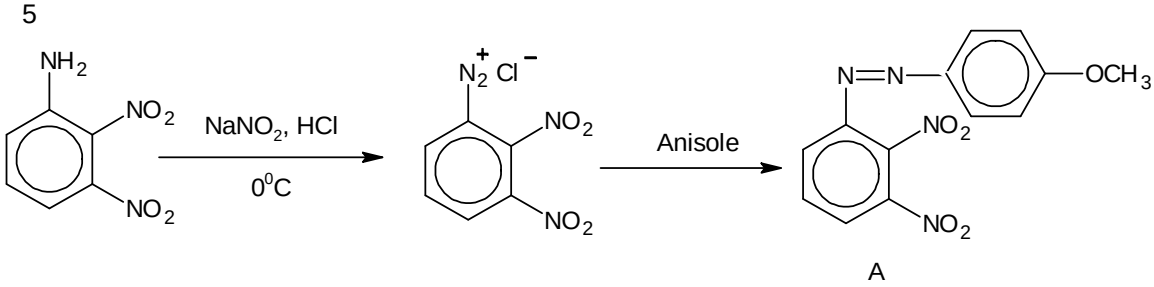
Degeneracy = 3

Statement – 3 In presence of Mg vicinal dihalides involved in elimination reaction.

Statement – 4 $[AgF_4]^-$ and $[AuCl_4]^-$ both are dsp^2 hybridized.

26. 5

Sol.



A

27. 3

Sol. The lower IP indicates that X will be alkali metal hen X-OH should be a hydroxide of alkali metal and higher value of IP Q indicates that It should be a halogen so QOH be an oxyacid. Hence the statement that pH of QH is more than 7 is incorrect.

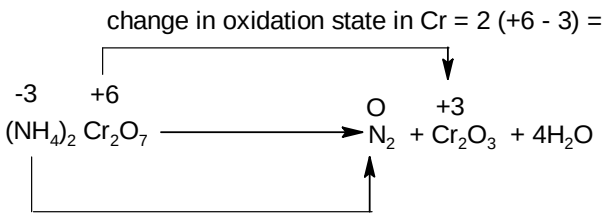
28. 3

Sol. $Xe \rightarrow Xe^+ + 1e^-$, I.E = 1170 kJmol^{-1}

$O_2 \rightarrow O_2^+ + 1e^-$, I.E = 1175 kJ mol^{-1}

29. 6

Sol.



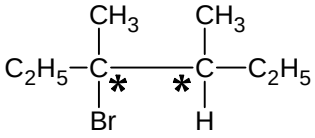
change in oxidation state = 2(-3-0)

In 'N" = - 6

Hence n_{factor} = 6

30. 4

Sol.



$= 2^2 = 4$

SECTION – C

31. 00009.65

Sol. $\Delta S^o = nF\left(\frac{dE^o}{dT}\right)_P$

$= 2 \times 96500C \times \left(5 \times 10^{-5} VK^{-1}\right)$

$= 9.65\ JK^{-1}$

32. 00001.69

Sol. $q = 0,\ w = -P\left(v_2 - v_1\right)$

$= -1(20-10)dm^3\ atm\ \ (\because\ 1\ atm\ dm^3 = 101.3\ J)$

$= -1013\ J$

$\Delta U = q + W \qquad (\text{from FLOT})$

$dU = -1013\ J$

$\Rightarrow \Delta U = nC_V\Delta T$ and $\Delta H = nC_P\Delta T$

$\Rightarrow \frac{\Delta H}{\Delta U} = \frac{C_P}{C_V} = \frac{(C_V + R)}{C_V}$

$= \frac{5/2R}{3/2R}$

$\Delta H = \frac{5}{3}\Delta U \Rightarrow \Delta H = \frac{5}{3} \times (-1013)$

$= -1688\ J$

33. 00034.89

Sol. $V = \pi \times 1 \times 1 \times 10 \times 10^3$ Litre

$= 3.14 \times 10^4$ Litre

$P = 0.8 \times 26 = 20.8$ mm of Hg

PV=nRT

$\frac{20.8}{760} \times 3.14 \times 10^4 = n \times 0.0821 \times 300 \Rightarrow n = 34.89$ mol

34. 00116.42

Sol. $P = x_T P_T^o + x_B P_T^o$ $\quad x_T, x_B$ mole fraction of Toluene and Benzene respectively.

= 67.525 mm of Hg = a

When last trace of liquid will disappear then $x_B\left(\text{vapour phase}\right) = \frac{1}{2},\ x_T\left(\text{vapour phase}\right) = \frac{1}{2}$

Then $\frac{1}{P} = \frac{x_T}{P_T^0} + \frac{x_B}{P_B^0}$

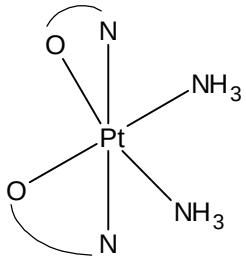
$$= \frac{0.5}{32.05} + \frac{0.5}{103}$$

P = 48.9 mm of Hg = b

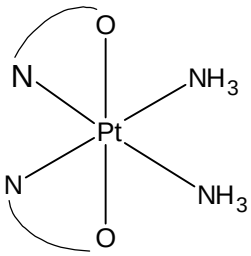
Then, $a + b = 116.42$ mm of Hg

35. 00052.50

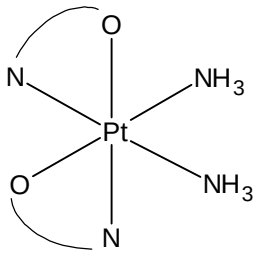
Sol.



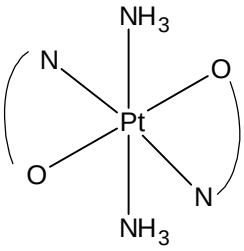
(i)



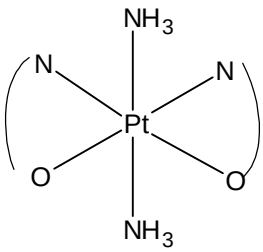
(ii)



(iii)



(iv)

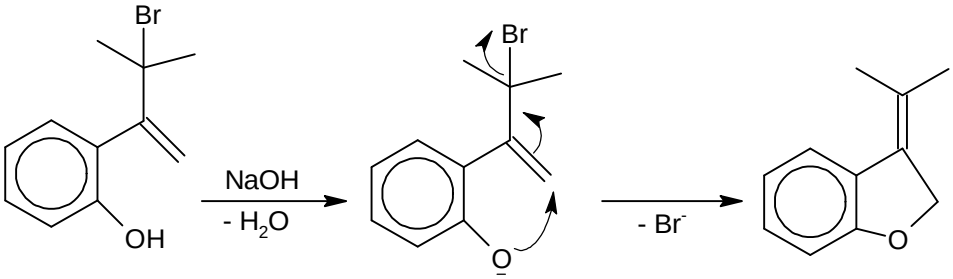


(v)

$\Rightarrow x = 5$ hence $\Rightarrow \frac{100 + 5}{2} = 52.5$

36. 00001.60

Sol.



M = 160

Mathematics

PART – III

SECTION – A

37. AD

Sol. Using family of lines $(a,b)=(1,1)$

$$\therefore \ell=\lim_{x\to 0^-}\frac{[\sin x]+\{\cos x\}}{\{x\}-1}=\frac{-2}{0}$$

38. AB

Sol. Using integration by parts

$$I_n=\left|x.\frac{1}{(1+x^2)^n}\right|_0^1-\int_0^1 x.\frac{(-n).2x}{(1+x^2)^{n+1}}dx$$

$$=\frac{1}{2^n}+2n\int_0^1\frac{x^2+1-1}{(1+x^2)^{n+1}}dx$$

$$=\frac{1}{2^n}+2n\,I_n-2n\,I_{n+1}$$

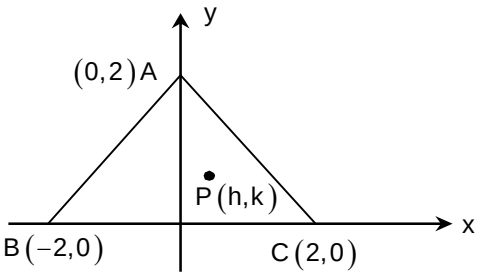
39. AC

Sol. $AB: y-x=2$

$AC: x+y=2$

$\therefore$ For P (h, k),

$$\left|\frac{h+k-2}{\sqrt{2}}\right|\left|\frac{h-k+2}{\sqrt{2}}\right|=2k^2$$



$$\frac{x^2}{16/3}+\frac{(y+2/3)^2}{(16/9)}=1 \qquad \therefore e=\sqrt{\frac{2}{3}}.$$

40. ABD

Sol. $\frac{dy}{dx}+y=f(x)\Rightarrow ye^x=\int e^x f(x)dx+c$

If $0\le x\le 2$, $ye^x=\int e^x.e^{-x}dx+c\Rightarrow ye^x=x+c$

If $x>2$, $ye^x=\int e^{x-2}dx\Rightarrow ye^x=e^{x-2}+c$

41. AD

Sol. $f_1(x)+f_2(x)+f_3(x)=0\Rightarrow D=-4a^2+b^2-4ac<0$

$f_1(x)+f_2(x)=0$ $D=(2a+b)^2-4(2b+c)\Rightarrow$ Nothing can be said.

$f_1+f_3=0$ $D=b^2-4ac-8a^2<0$

42. ABC

Sol. (A) No. of entries $=1+\dfrac{n(n+1)}{2}$

(B) No. of entries $=5+1$

(C) Minimum no. of zeros $=\dfrac{n(n-1)}{2}$

(D) Maximum no. of non-zero entire =10.

SECTION – B

43. 5

Sol. $\text{Area}(\Delta ABC)=\dfrac{1}{2}|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|$

Given $2\vec{a}+3\vec{b}+6\vec{c}\Rightarrow\vec{a}\times\vec{b}=2(\vec{c}\times\vec{a})$

$\Rightarrow 2(\vec{b}\times\vec{a})+6(\vec{b}\times\vec{c})\Rightarrow 3(\vec{b}\times\vec{c})=\vec{a}\times\vec{b}$

$$\therefore\frac{\text{Ar}(\Delta ABC)}{\text{Ar}(\Delta AOB)}=\frac{\frac{1}{2}|\vec{a}\times\vec{b}|\left(1+\frac{1}{2}+\frac{1}{3}\right)}{\frac{1}{2}|\vec{a}\times\vec{b}|}=\frac{11}{6}\therefore m=11,\ n=6$$

44. 7

Sol. $C_1:(x+1)^2+y^2=9$ $\quad C_2:(x-2)^2+y^2=49$

$CC_1=r+r_1$ $\quad CC_2=r_2-r\Rightarrow CC_1+CC_2=r_1+r_2=2a=10$

$C_1C_2=3=2ae\Rightarrow e=\dfrac{3}{10}$

45. 6

Sol. Put x =1 $f(1).f''(1)=(f'(1))^2$ Let $f'(1)=\alpha, f''(1)=\beta$

$\therefore\beta=\alpha^2$

Differentiate w.r.t x and put x = 1

$8+2\alpha\beta=2\alpha\beta+\alpha\beta\Rightarrow\alpha\beta=8$

$\therefore\alpha=2,\beta=4$

46. 3

Sol. $f(f(x))=4x+3$ and so on

$\therefore$ LHS $=30x+26$ $\quad\therefore x=3.$

47. 4

Sol. $\because\alpha+\beta=90^\circ$ $\quad\therefore\tan\beta=\cot\alpha$

$\tan\alpha+\cot\alpha\geq 2,\tan^2\alpha+\cot^2\alpha\geq 2$ $\quad\therefore\alpha=\beta=\pi/4.$

48. 7

Sol. $15^2+(15+d)^2+(15+2d)^2+.....+(15+9d)^2=1185$

On solving d = - 1

$s(n)\geq s(n-1)\Rightarrow n\leq 16 \qquad \therefore n_{max}=16$

SECTION – C

49. 00016.00

Sol. Put $x=1/t\Rightarrow dx=-1/t^2dt$

$$I=\int_{\infty}^{0}\frac{(t^2+at+1)}{1+t^4}\frac{t^4}{t^2}\cdot\left(\frac{-1}{t^2}\right)\tan^{-1}t\ dt$$

$$I=\int_{0}^{\infty}\frac{x^2+ax+1}{1+x^4}\tan^{-1}x\,dx$$

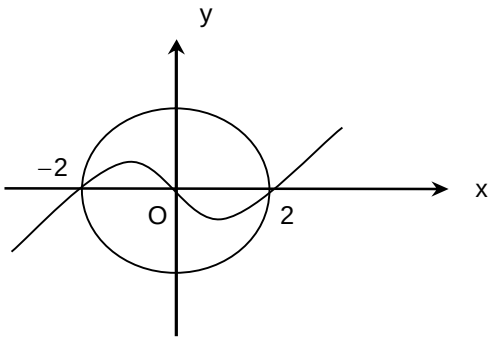
$$\therefore 2I=\int_{0}^{\infty}\frac{x^2+ax+1}{1+x^2}dx \quad \because \tan^{-1}x+\tan^{-1}\frac{1}{x}=\frac{\pi}{2}$$

$$\therefore I=\lim_{a\to\infty}\left(\frac{\pi^2}{8\sqrt{2a}}+\frac{\pi^2}{16}\right)\Rightarrow K=16$$

50. 00006.28

Sol. $f'(x)=a\left(x^2-\frac{4}{3}\right)\Rightarrow f(x)=\frac{2x}{3}(x^2-4)$

$\therefore$ Area bounded = 2π



51. 00003.40

Sol. $\because \sin^{-1}\left(\frac{2x}{1+x^2}\right)\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\therefore g(x)$ can be 0, 1, 2, 3

$f(g(x))<0\Rightarrow f(0)<0$ and $f(3)<0$

$$\therefore a\in\left(\frac{7}{5},2\right)$$

52. 00771.00

Sol. Case 1 – No box empty,

$x_1+x_2+x_3+x_4+x_5=10,\ x_i\geq 1\Rightarrow 126$

Case 2 – Exactly one empty, $= 420$

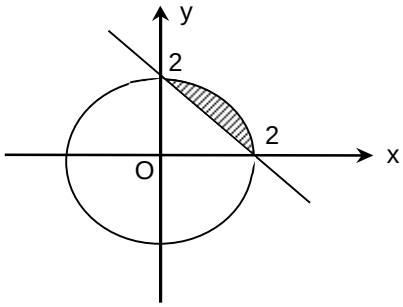
Case 3 – Exactly two empty = 216

Case 4 – Exactly three empty = 9

Total ways = 771

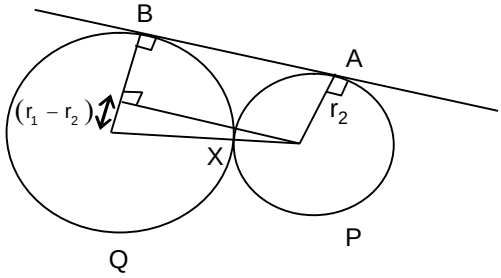
53. 00001.14

Sol. $\text{Area} = \frac{\pi.2^2}{4} - \frac{1}{2}.2.2 = \pi - 2$



54. 00011.11

Sol. $AB^2 = (r_1 + r_2)^2 - (r_1 - r_2)^2$



$= 2\sqrt{r_1 r_2}$

$r_1 = \frac{100}{9} \because r_2 = 9$